Unit Rates with Fractions
Answer Key

Grades 6–7 Ratios and Proportional Relationships

Quick Answers

1. 2
2. $\frac{3}{2}$
3. 4
4. 2
5. $\frac{12}{5}$, 2, —
6. $\frac{3}{4}$
7. $\frac{7}{4}$
8. 8
9. $\frac{8}{15}$
10. 3, 4, —
11. 6
12. 27

Curriculum

Level: Grades 6–7 Ratios and Proportional Relationships
6.RP.A.2, 6.RP.A.3 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12*
Difficulty 1–5 of 5 across 16 tagged checks.

Common wrong answers

12. *If they got 9: used the amount given as if it were already the rate per minute*

1. Zara jogs $\frac{1}{2}$ mile in $\frac{1}{4}$ hour — miles per hour?

Step 1. "Per hour" means the time has to become one hour, so divide the distance by the time: $\frac{1}{2} \div \frac{1}{4}$.
Step 2. Dividing by a fraction is multiplying by its reciprocal: $\frac{1}{2} \times \frac{4}{1} = \frac{4}{2}$.
Step 3. In lowest terms that is 2 miles per hour. $\boxed{2}$
Sense check: a quarter of an hour is four times a shorter stretch than an hour, so the distance should be four times bigger — and it is.

2. $\frac{3}{4}$ of a page in $\frac{1}{2}$ of a minute — pages per minute?

Step 1. The unit here is one minute, so divide the pages by the minutes: $\frac{3}{4} \div \frac{1}{2}$.
Step 2. Multiply by the reciprocal: $\frac{3}{4} \times \frac{2}{1} = \frac{6}{4}$.
Step 3. Reduce: $\frac{6}{4} = \frac{3}{2}$ pages per minute, which is one and a half pages. $\boxed{\frac{3}{2}}$
Listen for: a student who says the answer must be smaller than $\frac{3}{4}$ because it is a division. Dividing by a number below one makes things bigger.

3. $\frac{2}{3}$ **of a litre in** $\frac{1}{6}$ **of a minute — litres per minute?**

Step 1. Divide the litres by the minutes: $\frac{2}{3} \div \frac{1}{6}$.

Step 2. $\frac{2}{3} \times \frac{6}{1} = \frac{12}{3}$.

Step 3. That reduces to 4 litres per minute. $\boxed{4}$

Common wrong answer: $\frac{1}{4}$ — the division was written upside down, minutes over litres. Ask what the answer would mean: a quarter of a minute per litre is a different question.

4. $1\frac{1}{2}$ **cups of flour make** $\frac{3}{4}$ **of a batch — cups per batch?**

Step 1. One whole batch is the unit, so divide the flour by the part of a batch it made. First rewrite the mixed number: $1\frac{1}{2} = \frac{3}{2}$.

Step 2. $\frac{3}{2} \div \frac{3}{4} = \frac{3}{2} \times \frac{4}{3} = \frac{12}{6}$.

Step 3. That is 2 cups of flour for a whole batch. $\boxed{2}$

Teaching note: the mixed number must become improper before the reciprocal step. Flipping $1\frac{1}{2}$ as it stands is the classic slip.

5. Rowan $\frac{3}{5}$ **km in** $\frac{1}{4}$ **h; Sadie** $\frac{2}{3}$ **km in** $\frac{1}{3}$ **h.**

Step 1. The two walks cover different distances in different times, so nothing can be compared until both are written as kilometres per one hour.

Step 2. Rowan: $\frac{3}{5} \div \frac{1}{4} = \frac{3}{5} \times \frac{4}{1} = \frac{12}{5}$ km/h. $\boxed{\frac{12}{5}}$

Step 3. Sadie: $\frac{2}{3} \div \frac{1}{3} = \frac{2}{3} \times \frac{3}{1} = \frac{6}{3} = 2$ km/h. $\boxed{2}$

Step 4. $\frac{12}{5}$ is 2.4, and 2.4 is more than 2, so Rowan's speed is the greater of the two: $\boxed{>}$

Teaching note: Sadie covers more ground per trip, so a student who compares the distances alone picks the wrong walker. The rate is what settles it.

6. $\frac{5}{8}$ **of a mile in** $\frac{5}{6}$ **of an hour — miles per hour?**

Step 1. Divide the distance by the time: $\frac{5}{8} \div \frac{5}{6}$.

Step 2. $\frac{5}{8} \times \frac{6}{5} = \frac{30}{40}$.

Step 3. Reduce by 10: the unit rate is $\frac{3}{4}$ of a mile per hour. $\boxed{\frac{3}{4}}$

Common wrong answer: $\frac{4}{3}$ — the reciprocal was taken on the wrong fraction, giving hours per mile instead of miles per hour. A unit rate below the distance is fine here: the trip took most of an hour, so one full hour covers only a little more.

7. A hose fills $\frac{7}{8}$ **of a barrel in** $\frac{1}{2}$ **of an hour.**

Step 1. The question asks per hour, so divide barrels by hours: $\frac{7}{8} \div \frac{1}{2}$.

Step 2. $\frac{7}{8} \times \frac{2}{1} = \frac{14}{8}$.

Step 3. Reduce: $\frac{7}{4}$ of a barrel per hour, which is one and three quarters barrels. $\boxed{\frac{7}{4}}$

Sense check: half an hour filled most of a barrel, so a full hour should fill more than one. It does.

8. $\frac{3}{4}$ **of a pound of cheese costs** 6 **dollars — dollars per pound?**

Step 1. The unit is one pound, so divide the cost by the weight: $6 \div \frac{3}{4}$.

Step 2. A whole number divides the same way: $6 \times \frac{4}{3} = \frac{24}{3}$.

Step 3. That is 8 dollars for one pound. $\boxed{8}$

Sense check: three quarters of a pound cost six dollars, so a whole pound must cost more than six — eight is believable, four would not be.

9. Painting $\frac{2}{5}$ of a wall takes $\frac{3}{4}$ of an hour.
Step 1. Per hour means walls divided by hours: $\frac{2}{5} \div \frac{3}{4}$.
Step 2. $\frac{2}{5} \times \frac{4}{3} = \frac{8}{15}$, which will not reduce because 8 and 15 share no factor.
Step 3. In one hour, $\frac{8}{15}$ of the wall gets painted. $\boxed{\frac{8}{15}}$
Common wrong answer: $\frac{15}{8}$ — hours over wall instead of wall over hours. That number answers "how many hours for the whole wall", which is a fair question but not this one.

10. Brand A: $\frac{2}{3}$ lb for 2 dollars. Brand B: $\frac{3}{4}$ lb for 3 dollars.
Step 1. The packs are different sizes, so the sticker prices cannot be compared. Put both on the same footing: dollars for one pound.
Step 2. Brand A: $2 \div \frac{2}{3} = 2 \times \frac{3}{2} = 3$ dollars per pound. $\boxed{3}$
Step 3. Brand B: $3 \div \frac{3}{4} = 3 \times \frac{4}{3} = 4$ dollars per pound. $\boxed{4}$
Step 4. Three dollars a pound is less than four, so brand A is the better buy and the true statement uses the less-than sign: $\boxed{<}$
Teaching note: brand A is also the cheaper sticker price here, so ask the follow-up: what would happen if brand B's pack cost 2.50? The unit rate is what makes that answerable.

11. A cyclist rides $4\frac{1}{2}$ miles in $\frac{3}{4}$ of an hour.
Step 1. Rewrite the mixed number first: $4\frac{1}{2} = \frac{9}{2}$ miles. Then divide by the time.
Step 2. $\frac{9}{2} \div \frac{3}{4} = \frac{9}{2} \times \frac{4}{3} = \frac{36}{6}$.
Step 3. That is 6 miles per hour. $\boxed{6}$
Sense check: three quarters of an hour covered four and a half miles, so a full hour covers a bit more — six is right, and anything under four and a half would be wrong on sight.

12. $2\frac{1}{4}$ litres in $\frac{1}{3}$ of a minute — how many litres in 4 minutes?
Step 1. This is two steps. The machine's rate per minute is not given, so find the unit rate first, then multiply it by the number of minutes asked for.
Step 2. Unit rate: $2\frac{1}{4} = \frac{9}{4}$, and $\frac{9}{4} \div \frac{1}{3} = \frac{9}{4} \times \frac{3}{1} = \frac{27}{4}$ litres per minute.
Step 3. Now scale up: $\frac{27}{4} \times 4 = 27$ litres in four minutes. $\boxed{27}$
Common wrong answer: 9 — the given $2\frac{1}{4}$ litres was treated as the per-minute rate and multiplied by four. But that amount took only a third of a minute, so the rate is three times bigger.